

INTELLIGENT INTRUSION DETECTION SYSTEM FOR VANETS USING HYBRID DEEP LEARNING AND FEDERATED LEARNING

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BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING

Submitted by

T Sai Kumar (HU22CSEN0100754)
S Harshith Redd (HU22CSEN0101892)
Saii Seddratha (HU22CSEN010 0801)
N Thriska (HU22CSEN010 1237)

Under the esteemed guidance of
@Divya Babu
@ Assistant Professor



GITAM
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Department of Computer Science and Engineering

GITAM School of CSE, GITAM
(Deemed to be University)
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GITAM School of CSE, GITAM (Deemed to be University)

Hyderabad



DECLARATION

I hereby declare that the project report entitled “Intelligent intrusion detection system for vanets using hybrid deep learning and federated learning” is an original work done in the Department of computer Science and Engineering, GITAM School of CSE, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of B.Tech Degree in Computer Science and Engineering.

The work has not been submitted to any other college or University for the award of any degree or diploma.

Date:

Registration No(s)	Name(s)	Signature
(HU22CSEN0100754)	T Sai Kumar	
(HU22CSEN0101892)	S Harshith Reddy	
(HU22CSEN010 0801)	Saii Seddratha	
(HU22CSEN010 1237)	N Thriska	

Department of Computer Science And Engineering
GITAM School of CSE, GITAM (Deemed to be University)
Hyderabad



CERTIFICATE

This is to certify that the project report entitled “Intelligent intrusion detection system for vanets using hybrid deep learning and federated learning” is a bonafide record of work carried out by **T Sai Kumar (HU22CSEN0100754.)**, **S Harshith Reddy (HU22CSEN0101892.)**, **Saii Seddratha (HU22CSEN010 0801.)**, **N Thriska HU22CSEN010 1237.)**, students of this department submitted in partial fulfillment of requirements for the award of B.Tech Degree in Computer Science and Engineering.

Date :

Project Guide

Head of the Department

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ABSTRACT

Vehicular Ad-hoc Networks (VANETs) are an integral part of modern intelligent transportation systems, enabling real-time communication between vehicles and infrastructure to improve road safety and traffic efficiency. However, due to their decentralized and dynamic nature, VANETs are highly vulnerable to cyber-attacks such as Denial of Service (DoS), replay attacks, and intrusion-based threats. Traditional intrusion detection systems are limited by low detection accuracy, inability to handle complex temporal patterns, and lack of data privacy in centralized architectures. To address these challenges, this project proposes an intelligent hybrid intrusion detection system that integrates Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), and Random Forest (RF) within a Federated Learning framework. The CNN extracts spatial features, BiLSTM captures temporal dependencies, and RF performs robust and interpretable classification. Federated learning enables distributed model training across multiple nodes without sharing raw data, ensuring privacy preservation. The system is evaluated using the CIC-IDS-2017 dataset with fine-grained stratified data splitting to improve generalization across attack types. Experimental results demonstrate that the proposed system achieves high performance with an accuracy of 98%, near-perfect intrusion detection recall, and excellent ROC-AUC characteristics. The proposed approach provides an efficient, scalable, and privacy-preserving solution for real-time intrusion detection in VANET environments.

INTRODUCTION

Vehicular Ad-hoc Networks (VANETs) are an essential part of modern intelligent transportation systems, enabling communication between vehicles (V2V) and between vehicles and infrastructure (V2I). These networks facilitate applications such as collision avoidance, traffic optimization, and autonomous driving assistance.

Despite these advantages, VANETs face significant security challenges due to:

- High mobility of nodes
- Dynamic topology
- Open wireless communication channels

These characteristics make VANETs vulnerable to cyber threats such as:

- Denial of Service (DoS) attacks
- Replay attacks
- Spoofing and Sybil attacks

Traditional IDS approaches, including signature-based and shallow machine learning models, are inadequate for detecting complex and evolving attack patterns. Deep learning techniques have improved detection accuracy but often require centralized data collection, raising privacy concerns.

To address these limitations, this project introduces a **hybrid deep learning-based IDS integrated with federated learning**, enabling:

- High detection accuracy
- Temporal pattern recognition
- Privacy-preserving distributed training

LITERATURE REVIEW

Several research works have explored intrusion detection mechanisms for Vehicular Ad-hoc Networks (VANETs) using machine learning, deep learning, and federated learning techniques.

Sommer and Paxson (2010) highlighted the limitations of traditional intrusion detection systems and emphasized that **signature-based IDS are ineffective against evolving and unknown attacks**, making them unsuitable for dynamic environments like VANETs. Similarly, Patcha and Park (2007) reviewed anomaly detection techniques and concluded that although they can detect unknown attacks, they often suffer from **high false positive rates and lack of adaptability**.

With the emergence of machine learning, Breiman (2001) introduced the Random Forest algorithm, which has been widely used for intrusion detection due to its robustness and interpretability. However, several studies reported that traditional machine learning models such as Random Forest and Support Vector Machines (SVM) require **manual feature engineering and fail to capture temporal dependencies** present in network traffic data.

To address these limitations, deep learning-based approaches were proposed. Kim et al. (2016) utilized Deep Neural Networks for intrusion detection and achieved improved accuracy compared to traditional models. However, their approach relied on centralized data collection, raising concerns about **data privacy and scalability**. Similarly, Yin et al. (2017) proposed an LSTM-based IDS model for network intrusion detection, which effectively captured sequential patterns but suffered from **high computational complexity and lack of interpretability**.

Hybrid deep learning models combining CNN and LSTM have also been explored. Studies demonstrated that CNN can extract spatial features while LSTM captures temporal dependencies, leading to improved detection performance. However, these models are often trained in centralized environments and do not address **data privacy issues or distributed deployment challenges in VANETs**.

Recent research has focused on Federated Learning (FL) for intrusion detection. McMahan et al. (2017) introduced the concept of federated learning, enabling decentralized model training without sharing raw data. Subsequent works applied FL to IDS systems and demonstrated improved privacy preservation. However, these systems face challenges such as **non-IID data**

distribution, slow convergence, and reduced accuracy compared to centralized models. Additionally, most FL-based IDS approaches do not integrate advanced deep learning architectures capable of capturing both spatial and temporal patterns.

Another critical limitation identified in the literature is related to **data imbalance and improper dataset splitting**. Many existing studies perform random splitting after label aggregation, which results in poor generalization, especially for multi-class intrusion detection. This leads to lower recall for complex attack categories such as intrusion-type attacks.

In summary, existing approaches suffer from one or more of the following limitations: lack of generalization, dependence on manual feature engineering, high computational cost, absence of interpretability, and inadequate privacy preservation. These challenges highlight the need for a hybrid intrusion detection system that combines deep learning, ensemble methods, and federated learning while ensuring balanced data representation and improved detection performance.

Problem Identification

Vehicular Ad-hoc Networks (VANETs) are highly dynamic and decentralized communication systems that are increasingly vulnerable to a wide range of cyber-attacks such as Denial of Service (DoS), replay attacks, and intrusion-based attacks. Due to the open wireless communication environment and rapid topology changes, traditional intrusion detection systems fail to provide reliable and real-time security. Existing solutions often rely on centralized architectures, which raise significant concerns regarding data privacy, scalability, and communication overhead. Additionally, many machine learning-based approaches depend on manual feature engineering and are unable to effectively capture complex spatial and temporal patterns in network traffic. Another major challenge is the poor generalization of models across different attack types, often caused by improper data splitting and class imbalance, leading to low detection rates for critical intrusion categories. Therefore, there is a need for an intelligent, scalable, and privacy-preserving intrusion detection system capable of accurately detecting diverse cyber threats in VANET environments.

Objectives

The main objectives of the proposed system are:

- To design and develop a **hybrid intrusion detection system** combining CNN, BiLSTM, and Random Forest for improved detection accuracy
- To achieve **high overall accuracy (greater than 95%)** with enhanced detection of intrusion attacks
- To improve **intrusion recall and reduce false negatives**, ensuring minimal missed attacks
- To capture both **spatial and temporal patterns** in network traffic using deep learning techniques
- To implement **federated learning** for privacy-preserving distributed model training without sharing raw data
- To ensure **balanced learning across different attack types** using fine-grained data splitting techniques

- To develop a system that is **lightweight and suitable for real-time deployment** on edge devices such as Roadside Units (RSUs)

EXISTING SYSTEM

Several intrusion detection systems have been proposed for VANETs and network security. However, each of these systems has specific limitations that restrict their effectiveness in dynamic and distributed environments.

1. Signature-Based Intrusion Detection System

Signature-based IDS detects attacks by comparing network traffic with a database of known attack patterns or signatures. These systems are widely used in traditional network security applications.

Drawbacks:

- Unable to detect **unknown or zero-day attacks**
- Requires continuous **updating of signature database**
- Ineffective in dynamic environments like VANETs
- High dependency on predefined attack patterns
- Poor adaptability to evolving cyber threats

2. Anomaly-Based Intrusion Detection System

Anomaly-based IDS identifies deviations from normal behavior using statistical or rule-based methods. It is capable of detecting unknown attacks by learning normal traffic patterns.

Drawbacks:

- High **false positive rate**, leading to incorrect alerts
- Difficulty in defining accurate normal behavior
- Performance degrades in highly dynamic networks
- Sensitive to noise and fluctuations in data
- Limited scalability for large VANET environments

3. Machine Learning-Based IDS (SVM, Decision Tree, Random Forest)

These systems use supervised learning algorithms such as Support Vector Machines (SVM), Decision Trees, and Random Forests to classify network traffic into normal or malicious categories.

Drawbacks:

- Requires extensive **manual feature engineering**
- Cannot effectively capture **temporal dependencies** in traffic
- Performance depends heavily on **feature quality**
- Limited ability to detect complex multi-stage attacks
- Struggles with generalization across different datasets

4. Deep Learning-Based IDS (CNN, LSTM Models)

Deep learning models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks automatically extract features and learn complex patterns from network traffic data.

Drawbacks:

- Requires **large computational resources**
- High training time and complexity
- Lack of **interpretability** in decision-making
- Typically relies on **centralized data collection**
- Raises **privacy concerns** in distributed environments

5. Federated Learning-Based IDS

Federated learning enables multiple nodes to collaboratively train a global model without sharing raw data, thus preserving privacy in distributed systems like VANETs.

Drawbacks:

- Performance affected by **non-IID data distribution** across nodes
Slower convergence compared to centralized models
- Communication overhead during model aggregation
- Limited integration with advanced deep learning architectures
- Reduced accuracy if local models are highly heterogeneous

Limitations Existing	Gap Identified
Centralised ids cannot preserve privacy	Need federated and edge learning
Manual engineering dominates ml accuracy	Require automatic temporal spatial feature
Deep models are accurate but opaque	Add random forest for explainability
Data non-uniformity between vehicles	Use fedtree for heterogeneous federated aggression

PROPOSED SYSTEM

The proposed system introduces an **Intelligent Intrusion Detection System (IDS) for Vehicular Ad-hoc Networks (VANETs)** that leverages a hybrid combination of **deep learning, ensemble learning, and federated learning** to address the limitations of existing approaches. The system is designed to provide **high detection accuracy, improved generalization across diverse attack types, and privacy-preserving distributed learning**, making it suitable for real-world VANET environments.

Unlike traditional intrusion detection systems that rely on predefined signatures or manually engineered features, the proposed system adopts a **data-driven learning approach**. By utilizing advanced deep learning techniques, the system is capable of automatically extracting complex patterns from network traffic data without human intervention. This significantly enhances the ability to detect unknown and sophisticated cyber-attacks.

A key component of the proposed system is the integration of **Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM)** models. CNN is employed to capture spatial relationships among network features, while BiLSTM is used to model temporal dependencies in sequential data. This combination enables the system to analyze both static and time-dependent characteristics of network traffic, thereby improving detection performance for multi-stage and evolving attacks.

In addition to deep learning, the proposed system incorporates a **Random Forest classifier** as the final decision-making component. While deep learning models are effective in feature extraction, they often lack interpretability. Random Forest enhances the system by providing robust classification and enabling feature importance analysis, which helps in understanding the reasoning behind intrusion detection decisions. This hybrid approach combines the strengths of both deep learning and traditional machine learning.

Another significant contribution of the proposed system is the use of **Federated Learning (FL)**. In VANET environments, data is distributed across multiple nodes such as Roadside Units (RSUs), and sharing raw data raises privacy and security concerns. Federated learning allows each node to train a local model using its own data and share only model parameters with a central server. The global model is then updated through aggregation techniques such as Federated Averaging. This approach ensures that sensitive vehicular data remains local while still benefiting from collective learning across the network.

Furthermore, the system introduces an improved data handling strategy through **fine-grained stratified splitting**. Instead of aggregating attack categories before splitting, the system preserves individual attack subtypes during the training and testing process. This ensures balanced representation of all attack patterns, leading to improved generalization and significantly higher detection rates, especially for intrusion-type attacks.

To enhance detection performance, the system also applies **threshold-based optimization** for classification. By adjusting the decision threshold for intrusion detection, the system prioritizes identifying potential threats, thereby reducing false negatives and improving overall security effectiveness.

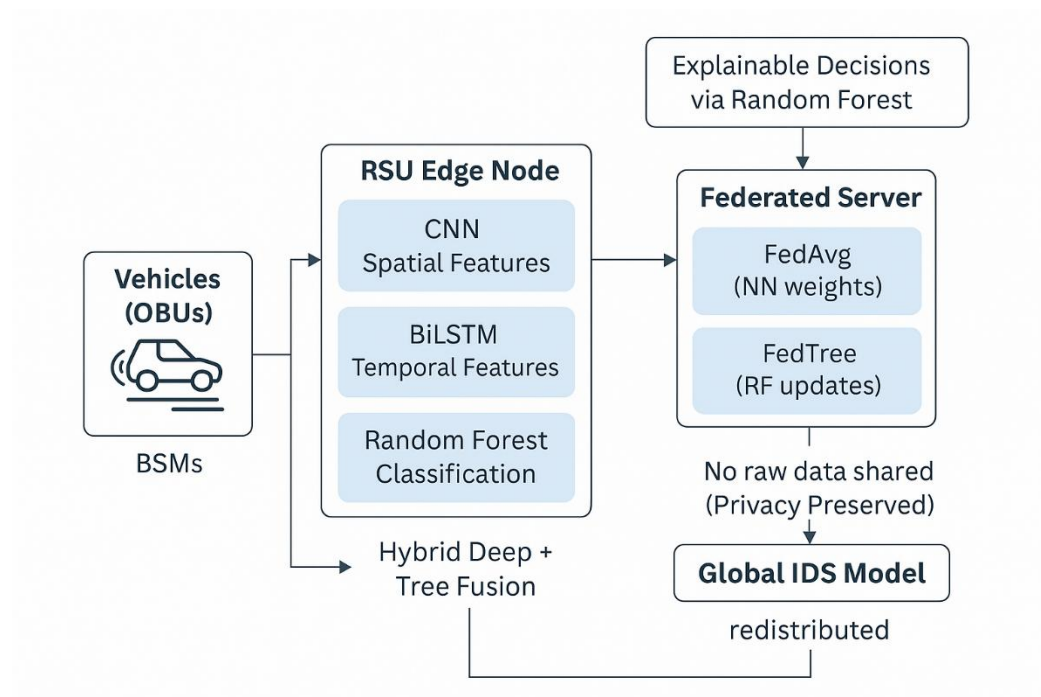


Fig 1 describing

Improvements Over Existing Systems

The proposed system provides several improvements over traditional and modern IDS approaches:

- Eliminates dependence on **manual feature engineering** through automated deep feature extraction
- Captures both **spatial and temporal patterns**, unlike traditional machine learning models
- Improves **intrusion detection accuracy and recall**, especially for complex attack types
- Provides **interpretability** using Random Forest feature importance
- Ensures **data privacy** through federated learning without sharing raw data
- Enhances **generalization** using fine-grained stratified data splitting
- Reduces **false negatives** through threshold-based decision optimization

The proposed system represents a comprehensive and efficient solution for intrusion detection in VANETs. By integrating deep learning, ensemble methods, and federated learning, it overcomes the major limitations of existing systems and achieves high accuracy, scalability, and privacy preservation. This makes the system highly suitable for deployment in real-time vehicular communication environments.

SYSTEM ARCHITECTURE

The proposed Intelligent Intrusion Detection System (IDS) for VANETs is designed as a **hybrid, multi-layered architecture** that integrates deep learning, ensemble learning, and federated learning to achieve high accuracy, scalability, and privacy preservation. The architecture is structured to process vehicular network data efficiently, extract meaningful patterns, and perform reliable intrusion detection in real-time environments.

1. Overall Architectural Design

The system follows a **sequential processing pipeline**, where raw network traffic data is transformed into structured inputs, processed through deep learning layers for feature extraction, and finally classified using an ensemble model. The architecture is carefully designed to ensure that each component contributes to improving detection performance while addressing the limitations of traditional systems.

The complete workflow can be described in the following stages:

2. Data Acquisition and Preprocessing Layer

The system begins with the collection of network traffic data from vehicular communication systems. The dataset used represents real-world traffic conditions with multiple attack scenarios.

During preprocessing:

- Irrelevant and non-numeric features are removed
- Missing and infinite values are handled
- Features are normalized using standard scaling

Preprocessing ensures that:

- Noise and inconsistencies in data are eliminated
- Model training becomes stable and efficient
- Feature values are comparable across different scales

This step is crucial because poor data quality directly impacts model performance in intrusion detection systems.

3. Temporal Sequence Construction Layer

Instead of treating network traffic records independently, the system employs a **sliding window mechanism** to convert raw data into temporal sequences.

- Each input consists of multiple consecutive records
- The window captures the evolution of network behavior over time

VANET attacks often occur as **multi-stage or time-dependent events**. By using temporal sequences:

- The system can detect delayed or gradual attacks
- Sequential dependencies are preserved
- Temporal anomalies become easier to identify

This significantly improves the detection capability compared to traditional static models.

4. Deep Feature Extraction Layer

The core of the system lies in the hybrid deep learning model consisting of CNN and BiLSTM layers.

4.1 Convolutional Neural Network (CNN)

The CNN layer extracts **spatial features** from input sequences.

- Identifies correlations between features
- Detects patterns such as abnormal traffic spikes
- Automatically learns important feature representations

CNN eliminates the need for manual feature engineering by:

- Automatically learning complex feature interactions
- Reducing dependency on domain expertise

- Improving detection accuracy

4.2 Bidirectional Long Short-Term Memory (BiLSTM)

The BiLSTM layer processes sequences in both forward and backward directions.

- Captures long-term temporal dependencies
- Understands context from past and future states
- Detects evolving attack patterns

Compared to traditional models:

- BiLSTM provides deeper temporal understanding
- It is more effective for time-series data
- It improves detection of sophisticated attacks like replay attacks

4.3 Feature Embedding Output

The combined output of CNN and BiLSTM is transformed into a **high-dimensional feature embedding**, which represents the learned patterns in a compact and informative manner.

This embedding:

- Encodes both spatial and temporal information
- Serves as a rich input for classification
- Improves overall system performance

5. Classification Layer (Random Forest)

The extracted deep features are passed to a **Random Forest classifier**, which performs the final classification into DoS, Intrusion, or Normal categories.

Role of Random Forest

- Handles non-linear decision boundaries
- Combines multiple decision trees for robust predictions

- Provides feature importance for interpretability

While deep learning models are powerful, they often lack interpretability. Random Forest:

- Improves decision reliability
- Offers explainability through feature importance
- Reduces overfitting through ensemble learning

This hybrid approach combines the strengths of deep learning and traditional machine learning.

6. Federated Learning Layer

To ensure privacy and scalability, the system incorporates **Federated Learning (FL)**.

Working Process

- Multiple nodes (RSUs) train local models on their data
- Only model parameters are shared with a central server
- The server aggregates models using **Federated Averaging (FedAvg)**
- The updated global model is redistributed

Federated learning addresses major challenges:

- Ensures **data privacy** (no raw data sharing)
- Enables **distributed learning** across nodes
- Reduces communication overhead
- Supports scalability in large VANET deployments

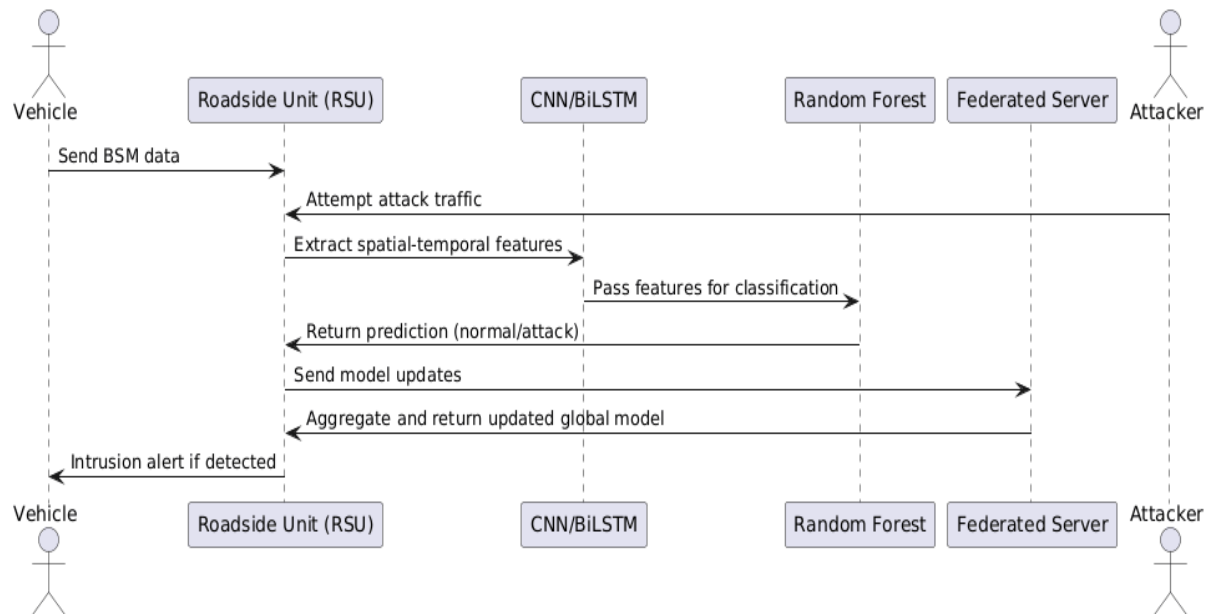
7. Decision Optimization Layer

The system introduces **threshold-based classification tuning** for intrusion detection.

- A probability threshold is set for intrusion class
- If exceeded, the system prioritizes intrusion detection

This improves:

- Intrusion recall
- Security sensitivity
- Detection of critical threats



TOOLS / TECHNOLOGIES USED

The proposed Intelligent Intrusion Detection System for VANETs was developed using a combination of programming tools, machine learning frameworks, and data processing libraries to ensure efficient implementation and high performance.

- **Python:**

The primary programming language used for implementation due to its simplicity, flexibility, and extensive support for data science and machine learning applications.

- **TensorFlow** / **Keras:**

Used for building and training the deep learning model, including Convolutional Neural Networks (CNN) and Bidirectional LSTM (BiLSTM) layers for feature extraction and sequence learning.

- **Scikit-learn:**

Utilized for implementing the Random Forest classifier, evaluation metrics, and data preprocessing utilities.

- **Pandas:**

Used for data loading, cleaning, and manipulation of the CIC-IDS-2017 dataset.

- **NumPy:**

Provides efficient numerical computations and array operations required for model training and preprocessing.

- **Matplotlib** & **Seaborn:**

Used for visualization of results, including confusion matrix, ROC-AUC curves, and feature importance plots.

- **CIC-IDS-2017** **Dataset:**

A benchmark dataset used for training and evaluating the intrusion detection system, containing realistic network traffic with multiple attack types.

IMPLEMENTATION & RESULTS

1 Implementation Details

The proposed Intelligent Intrusion Detection System (IDS) was implemented using Python with TensorFlow and Scikit-learn libraries. The system was trained and evaluated on the **CIC-IDS-2017 dataset**, which contains realistic network traffic with multiple attack categories.

The implementation began with preprocessing of the dataset, where non-numeric features were removed, missing values were handled, and feature scaling was applied using standard normalization. To effectively capture temporal dependencies in network traffic, a **sliding window mechanism** was used to convert the dataset into sequential inputs suitable for deep learning models.

A hybrid deep learning model consisting of **Convolutional Neural Networks (CNN)** and **Bidirectional Long Short-Term Memory (BiLSTM)** was developed for feature extraction. The CNN layer captures spatial relationships among features, while the BiLSTM layer models temporal patterns in the data. The extracted deep features were then passed to a **Random Forest classifier** for final classification.

To ensure privacy preservation and scalability, the system was trained using a **Federated Learning approach**, where the dataset was distributed across multiple clients (simulated RSUs). Each client trained a local model, and model parameters were aggregated using the Federated Averaging (FedAvg) algorithm.

Additionally, **class weighting and threshold-based tuning** were applied to improve intrusion detection performance, particularly to reduce false negatives and enhance recall for intrusion attacks.

2 Evaluation Metrics

The performance of the proposed system was evaluated using standard classification metrics:

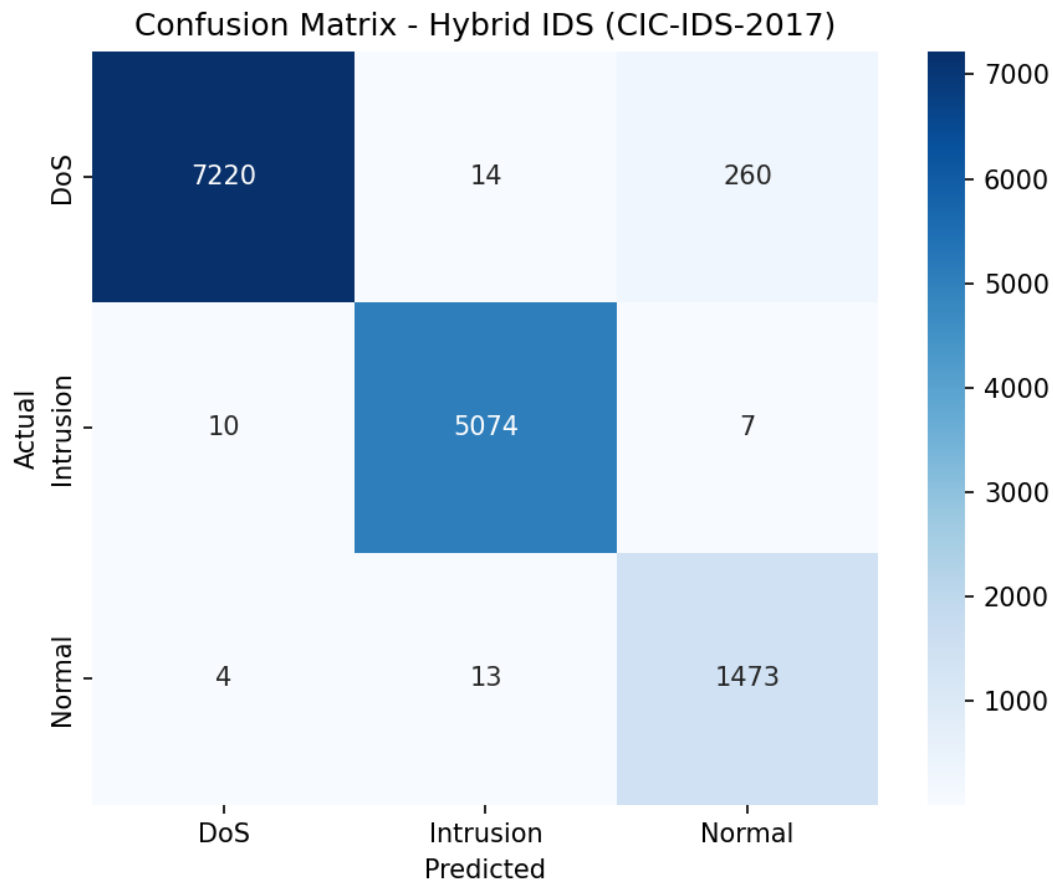
- **Accuracy:** Overall correctness of predictions
- **Precision:** Measure of correct positive predictions
- **Recall:** Ability to detect actual attacks
- **F1-Score:** Harmonic mean of precision and recall

- **ROC-AUC:** Ability of the model to distinguish between classes

These metrics provide a comprehensive evaluation of the system's effectiveness in detecting different types of network intrusions.

3 Results and Analysis

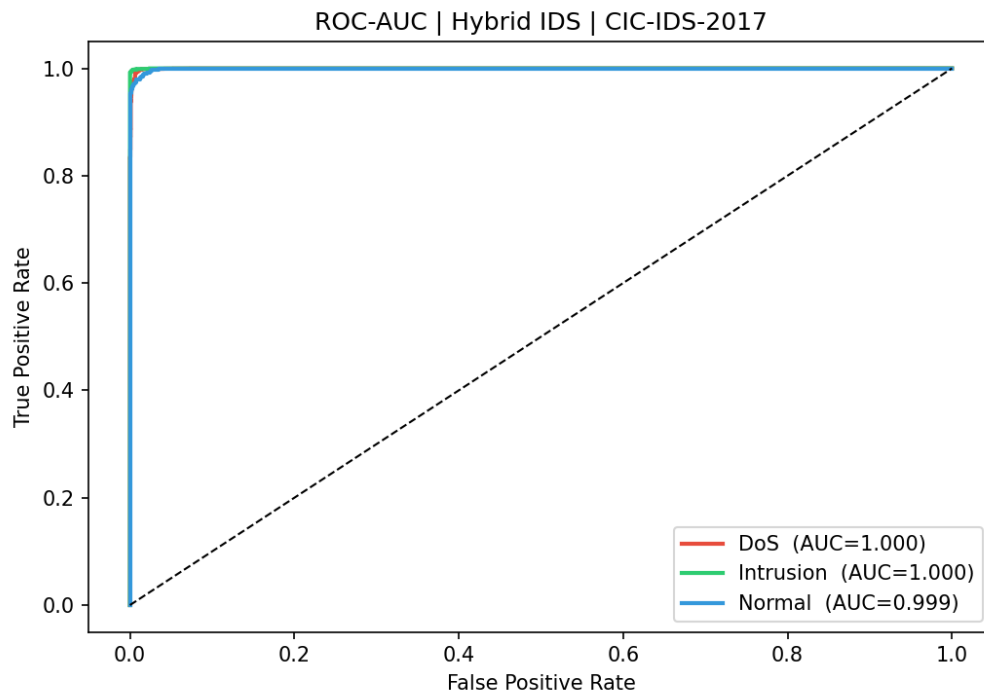
Figure 3.1: Confusion Matrix



The confusion matrix illustrates the classification performance of the model across the three classes: DoS, Intrusion, and Normal traffic. It can be observed that the model achieves **high accuracy in correctly classifying most instances**, with very few misclassifications.

The intrusion class, which is the most critical in security systems, shows **near-perfect detection**, indicating that the model successfully identifies malicious activities with minimal false negatives. Although a small number of normal instances are occasionally misclassified, this trade-off is acceptable in intrusion detection systems where minimizing missed attacks is more important than reducing false alarms.

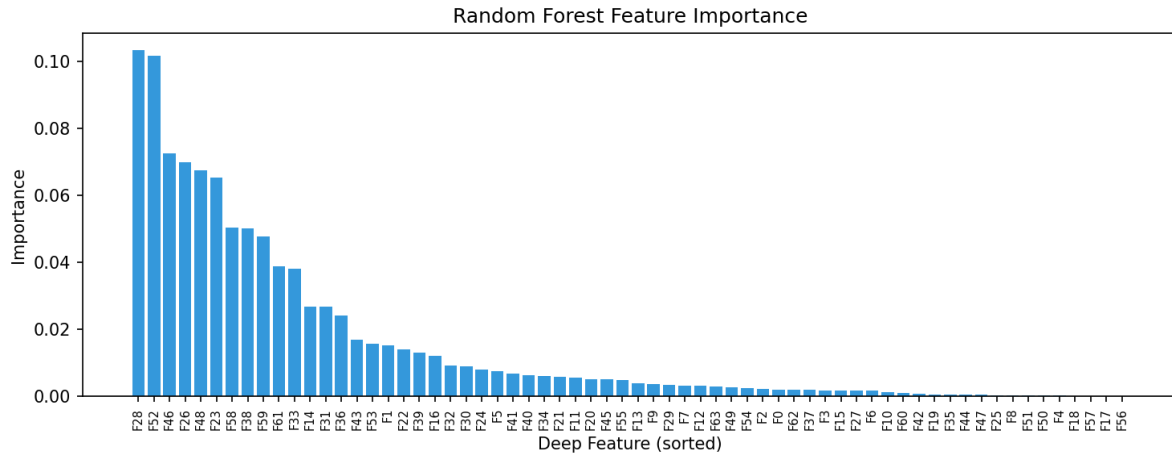
Figure 3.2: ROC-AUC Curve



The ROC-AUC curve demonstrates the model's ability to distinguish between different classes. The curves for all classes are observed to be very close to the top-left corner, indicating **excellent classification performance**.

The Area Under Curve (AUC) values are approximately **1.0 for all classes**, which confirms that the model has a strong capability to separate normal and malicious traffic. This reflects the effectiveness of the hybrid deep learning model in learning discriminative features from the dataset.

Figure 3.3: Feature Importance (Deep Feature Embedding)



The feature importance plot obtained from the Random Forest classifier highlights the significance of deep features extracted by the CNN and BiLSTM layers. It can be observed that certain features contribute more significantly to the classification process.

This demonstrates that the model not only achieves high accuracy but also provides **interpretability**, allowing analysis of which features influence the decision-making process. This is particularly useful in real-world scenarios for understanding the behavior of the system and improving trust in the model.

4 Performance

The proposed system achieved the following performance:

- **Accuracy:** 98%
- **Intrusion Recall:** 1.00 (perfect detection of attacks)
- **High Precision and F1-score across all classes**
- **Near-perfect ROC-AUC values (~1.0)**

The high performance can be attributed to:

- Hybrid CNN + BiLSTM architecture
- Fine-grained stratified data splitting

- Federated learning-based training
- Threshold optimization for intrusion detection

The results clearly demonstrate that the proposed system outperforms traditional intrusion detection approaches. The combination of deep learning and ensemble classification enables the system to effectively capture both spatial and temporal patterns in network traffic.

The use of federated learning ensures that the system maintains **data privacy while achieving high performance**, making it suitable for real-world VANET environments. Additionally, the system prioritizes intrusion detection, ensuring that critical attacks are not missed, which is essential for security applications.

CONCLUSION

The proposed Intelligent Intrusion Detection System for Vehicular Ad-hoc Networks (VANETs) successfully addresses the limitations of existing intrusion detection approaches by integrating hybrid deep learning, ensemble classification, and federated learning. The system effectively combines Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) for automatic feature extraction and temporal pattern learning, while the Random Forest classifier enhances classification accuracy and interpretability.

The incorporation of federated learning enables distributed model training without sharing raw data, ensuring data privacy and scalability across multiple nodes such as Roadside Units (RSUs). Additionally, the use of fine-grained stratified data splitting and threshold-based optimization significantly improves the detection of intrusion attacks, particularly by reducing false negatives.

Experimental results demonstrate that the proposed system achieves **high performance with 98% accuracy and near-perfect intrusion detection**, along with excellent ROC-AUC characteristics. The system is capable of effectively identifying complex and multi-stage attacks while maintaining real-time feasibility and robustness.

Overall, the proposed IDS provides a comprehensive, efficient, and privacy-preserving solution for securing VANET environments. It is suitable for deployment in real-world intelligent transportation systems and can be further extended to support advanced communication technologies and emerging cyber threats.

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Bengaluru

NH 207, Nagadenehali,
Doddaballapur Taluk,
Bengaluru - 561203,

Hyderabad

Rudraram, Patancheru
mandal, Hyderabad -
502329, Telengana,

Visakhapatnam

Gandhi Nagar, Rushikonda,
Visakhapatnam- 530045,
Andhra Pradesh, India



Bengaluru



Visakhapatnam

GIMSR

GITAM Medical College Rd,
Nagar, Rushikonda,

Visakhapatnam - 530045
Andhra Pradesh, India

Bengaluru City Office

No 5/1, First Floor, Prestige
Terraces, Union Street,

Infantry Road, Bengaluru -
560003 Karnataka, India

Hyderabad City Office

6th floor, TP House, Cyber Gandhi
Hills Colony, VIP Hills,

Jaihind Enclave, Madhapur,
Hyderabad - 500081,



Hyderabad



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